

Hospital Investment Case

Prepared by: Stella Obase (STELLANALYZES)

Program: Dataverse Africa — Healthcare Data Analyst Track, Cohort 4.0

Analysis period: January 2022 – December 2023 (24 months of encounter-level and claims data)

Candidates evaluated: Greenfield General Hospital (GGH), Eastpoint Community Hospital (ECH), Riverside Medical Centre (RMC)

Executive Summary

NorthBridge Health Investment Fund commissioned this independent analysis to determine which of three shortlisted hospitals; Greenfield General Hospital (GGH), Eastpoint Community Hospital (ECH), and Riverside Medical Centre (RMC); should receive a five-year capital investment to expand capacity, modernize operations, and improve service delivery.

Two years of encounter-level and claims data per hospital were cleaned, modeled, and scored across five weighted investment pillars: Financial Performance (22%), Clinical Quality (30%), Claims Efficiency (18%), Operational Efficiency (18%), and Growth & Scalability (12%). Each hospital was scored 1–10 on 20 criteria, and weighted scores were summed into a single composite investment score out of 10.

Recommendation: NorthBridge should invest in Greenfield General Hospital (GGH). GGH earned the highest weighted composite score (6.30 / 10, vs. 6.09 for ECH and 5.93 for RMC), driven by clinical-quality leadership and genuine, capacity-backed room to scale. Its financial and claims-efficiency scores are the field's weakest, but both trace to a fixable revenue-cycle-management gap rather than a structural or demand problem.

Background & Objective

NorthBridge Health Investment Fund is a private equity firm focused on healthcare infrastructure across sub-Saharan Africa. The fund has shortlisted three hospitals for a major capital investment; only one will receive funding. As the engaged analyst, the objective was to evaluate each hospital's operational performance, financial sustainability, clinical quality, and growth potential, and recommend the single strongest investment candidate — supported directly by the underlying data, not by reputation or assumption.

Methodology

Each hospital was scored 1–10 on 20 criteria across five weighted pillars, using its relative standing against the other two candidates on the underlying KPI (the strongest performer on a given metric score highest). Weighted Score = Weight × Raw Score; the sum of all 20 weighted scores produces each hospital's Weighted Total Score out of 10.

Scoring Approach

- Each hospital is scored 1–10 (10 = strongest) on 20 criteria grouped into 5 investment pillars.
- Scores were assigned by ranking the three hospitals' underlying KPI values and translating relative standing into a 1–10 scale; scores are not on an absolute clinical or industry benchmark.

Investment Pillars & Weights

Pillar	Weight	What it measures & why it matters
Financial Performance	22%	Revenue base, collection strength, and billing-to-cash leakage — returns depend on cash actually collected, not billed.
Clinical Quality	30%	Mortality, infection, readmission, complication, recovery, and satisfaction — the largest weight; the most durable, hardest-to-replicate value driver.
Claims Efficiency	18%	Denial rate, revenue at risk from denials, and claims processing speed — a proxy for revenue-cycle maturity and near-term fixability.
Operational Efficiency	18%	ED wait time, encounters per bed per year, and average length of stay — throughput and capacity utilization that drive unit economics.
Growth & Scalability	12%	Revenue and encounter YoY growth plus Capacity Headroom (inverse of enc/bed/yr) — whether a hospital can actually convert growth into scale, not just whether revenue moved last year.

Data & Scope

- Encounter-level source data:
- East Community Hospital – 3,500 encounters
- Greenfield General Hospital – 3,600 encounters
- Riverside Medical Centre – 3,800 encounters

Findings by Pillar

Financial Performance (22%)

Riverside Medical Centre (RMC) generates the strongest revenue base; the highest average revenue per encounter and the highest collection rate of the three hospitals. Greenfield (GGH) has the smallest revenue base and the highest revenue leakage, but this reflects a claims-and-collections gap rather than weak patient demand (GGH's encounter volume and clinical outcomes are strong).

Criterion	Weight	GGH	ECH	RMC
Total Revenue	7%	3	5	8
Collection Rate	8%	4	6	8
Revenue Leakage	7%	4	6	6

Clinical Quality (30%)

GGH leads on every clinical-quality metric measured: the lowest mortality rate, the lowest hospital-acquired infection rate, the lowest readmission rate, the lowest complication rate, and the highest recovery rate of the three hospitals. RMC, despite its financial strength, posts the weakest results across this entire pillar; the highest mortality, complication, and readmission rates, and the lowest patient satisfaction score. This pillar carries the largest weight of the five, reflecting its role as the most durable, hardest-to-replicate driver of long-run value.

Criterion	Weight	GGH	ECH	RMC
Mortality Rate	6%	9	7	4
HAI Rate	5%	9	7	5
Readmission Rate	5%	9	6	3
Complication Rate	5%	8	7	4
Recovery Rate	5%	9	8	6
Patient Satisfaction	4%	7	7	5

Claims Efficiency (18%)

RMC's revenue-cycle operation is the most mature in the field: the lowest denial rate and the least billed revenue at risk from denials. GGH's denial rate is the highest of the three hospitals and is the single largest identifiable lever for improving its financial position without any change in patient volume or clinical performance.

Criterion	Weight	GGH	ECH	RMC
Denial Rate	7%	3	6	9
Denied Revenue at Risk	6%	3	5	8
Avg Processing Days	5%	8	7	6

Operational Efficiency (18%)

GGH has the shortest emergency-department wait time and shortest average length of stay of the three candidates. ECH posts the highest raw encounters-per-bed figure, but Section 5.5 shows this reflects near-full capacity utilization rather than superior efficiency — ECH has little room left to absorb further growth.

Criterion	Weight	GGH	ECH	RMC
Avg ED Wait Time	6%	8	6	4
Encounters per Bed/Yr	6%	6	9	4
Avg LOS	6%	8	6	5

Growth & Scalability (12%) — revised methodology

This scoring uses two signals: raw year-over-year revenue and encounter growth, and Capacity Headroom (the inverse of encounters per bed per year), which measures how much room each hospital has to scale before it needs new capital investment in beds.

ECH posts the strongest headline revenue growth (+5.7% YoY), but its encounter volume is falling (-5.0%) and its encounters/bed/year (9.21) leaves almost no spare capacity. It cannot scale further without significant capital expenditure on new beds, and its revenue gain may simply reflect rising per-encounter pricing on a shrinking patient base. GGH's revenue declined (-18.5%) and encounter volume softened modestly (-6.2%), but its encounters/bed/year (6.43) sits at a healthy mid-range; genuine room to scale once revenue-cycle reform unlocks reinvestment capital, consistent with the turnaround thesis already established in the Financial and Claims Efficiency pillars. RMC has the most headroom (3.96 enc/bed/year) but this reflects chronic underutilization driven by its weaker clinical and operational profile, not unmet scalable demand.

Criterion	Weight	GGH	ECH	RMC
Revenue YoY Growth	5%	6	6	4
Encounter Volume YoY Growth	4%	6	2	7
Capacity Headroom (Scalability)	3%	9	1	10

Weighted Investment Scorecard

The table below summarizes the full 20-criterion scorecard. Full detail, including live formulas, is provided in the accompanying Excel workbook.

Pillar (Weight)	GGH	ECH	RMC
Financial Performance (22%)	0.81	1.25	1.62
Clinical Quality (30%)	2.57	2.10	1.34
Claims Efficiency (18%)	0.79	1.07	1.41
Operational Efficiency (18%)	1.32	1.26	0.78
Growth & Scalability (12%)	0.81	0.41	0.78
WEIGHTED TOTAL SCORE (of 10)	6.30	6.09	5.93

Risks & Considerations

Greenfield General Hospital (GGH) — recommended

- Revenue and denial rate are the weakest in the field, but both are addressable through a revenue-cycle-management engagement rather than a structural or demand issue.
- Smallest current revenue base means a slower absolute payback period even after collections improve.

Eastpoint Community Hospital (ECH)

- Posts the strongest headline revenue growth of the three hospitals, but is running near-max bed capacity (9.21 encounters/bed/year). It cannot scale further without new capital spend on beds.
- Mid-pack performance on clinical quality limits upside potential versus the recommended candidate even before the capacity constraint is considered.

Riverside Medical Centre (RMC)

- The strongest financial and claims performance of the three, but the weakest clinical-quality and patient-satisfaction scores.
- Elevated mortality, complication, and readmission rates carry regulatory and reputational risk that could offset financial strength over a five-year holding period.
- Its high-capacity headroom reflects chronic underutilization, not unmet scalable demand.

Recommendation

NorthBridge should invest in Greenfield General Hospital. GGH's clinical-quality leadership and genuine, capacity-backed room to scale represent the strongest foundation for a five-year value-creation plan, and its principal weakness; a high claims-denial rate and associated revenue leakage is a known, fixable revenue-cycle problem rather than a signal of poor demand, poor care, or poor management. ECH's revenue growth is real but is structurally capped by near-full bed utilization; RMC's financial strength is real but is offset by materially weaker patient-safety outcomes that pose ongoing regulatory and reputational risk.

Suggested 90-day path to close:

Weeks 1–4: confirmatory due diligence on GGH's revenue-cycle operations and its top denial-reason categories.

Weeks 5–8: engage a revenue-cycle-management partner to design a denial-reduction and collections work plan.

Weeks 9–12: finalize investment terms, capacity-expansion scope, and a 12-month performance covenant tied to collection rate and denial rate.

Supporting Files

- Excel workbook: Northbridge_Hospital_Investment_Scorecard.xlsx — full scorecard, and underlying KPI evidence.
- PowerPoint deck: Northbridge_Investment_Committee_Deck.pptx; executive presentation for the investment committee.
- SQL: NORTHBRIDGE_CAPSTONE.sql — data cleaning, KPI aggregation, and scorecard export queries.
- Power BI: Northbridge_investment_project.pbix — interactive executive dashboard.